

Policy-Based Methods



Outline

- **Policy Optimization and Policy Gradient**



Reinforcement Learning

- **Model-based RL**

- Learn the reward function and transition probabilities
- Use planning (e.g., Value Iteration)
- Extract policy

- **Value-based RL**

- Learn the value function directly (e.g., Q^*)
- Extract policy

- **Policy-based RL**

- Learn the policy directly



Policy Optimization

- Policy-based methods are also model-free methods
- Methods that represent a policy explicitly $\pi_{\theta}(a|s)$
- These methods optimize the parameters θ either:
 - directly by gradient ascent on the performance objective $J(\pi_{\theta})$,
 - or indirectly, by maximizing local approximations of $J(\pi_{\theta})$

Policy Gradient

We aim to maximize the expected return

$$J(\pi_{\theta}) = E_{\tau \sim \pi_{\theta}}[R(\tau)]$$

Where:

π_{θ} - stochastic and parameterized policy

τ – trajectory $(s_0, a_0, s_1, a_1, \dots, s_T)$

$R(\tau)$ - finite-horizon undiscounted return*

* the derivation for the infinite-horizon discounted return setting is almost identical

Policy Gradient

We will optimize the policy by gradient ascent:

$$\theta_{k+1} = \theta_k + \alpha \nabla_{\theta} J(\pi_{\theta})|_{\theta_k}$$

Where:

$\nabla_{\theta} J(\pi_{\theta})$ is called the **policy gradient**

Algorithms that optimize the policy with this method are called **policy gradient algorithms**

Policy Gradient

To use policy gradient algorithms, we follow these steps:

- **Derive the gradient**
- **Compute a sample estimate of the expected value**, which can be computed with data from a finite number of agent-environment interaction steps

Derivation and code from **Open AI Spinning Up**:

https://spinningup.openai.com/en/latest/spinningup/rl_intro3.html

Policy Gradient

- A few interesting facts that are useful for deriving the gradient analytically
 - **Probability of a trajectory**
 - The Log-Derivative Trick
 - Log-Probability of a Trajectory
 - Gradients of Environment Functions
 - Grad-Log-Prob of a Trajectory

Policy Gradient

- **Probability of a Trajectory**

- The probability of a trajectory $\tau = (s_0, a_0, s_1, a_1, \dots, s_{T+1})$ given that actions come from π_θ is

$$P(\tau|\theta) = \rho_0(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) \pi_\theta(a_t|s_t)$$

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Policy Gradient

- **The Log-Derivative Trick**

- The log-derivative trick is based on a simple rule from calculus:

- the derivative of $\log(x)$ with respect to x is:

$$\frac{d}{dx} \log(x) = \frac{1}{x}$$

- Hence:

$$\frac{d}{dx} f(x) = f(x) \frac{d}{dx} \log(f(x))$$

Policy Gradient

- The Log-Derivative Trick

- Hence:

$$\nabla_{\theta} P(\tau|\theta) = P(\tau|\theta) \nabla_{\theta} \log(P(\tau|\theta))$$

Policy Gradient

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Policy Gradient

- **Log-Probability of a Trajectory**

- Recall the probability of a trajectory:

$$P(\tau|\theta) = \rho_0(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) \pi_{\theta}(a_t|s_t)$$

- Hence the log-prob of a trajectory is:

$$\begin{aligned} & \log(P(\tau|\theta)) \\ &= \log(\rho_0(s_0)) + \sum_{t=0}^T (\log(P(s_{t+1}|s_t, a_t)) + \log(\pi_{\theta}(a_t|s_t))) \end{aligned}$$

Policy Gradient

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Policy Gradient

- **Gradients of Environment Functions**
 - The environment has no dependence on θ
 - Hence, the **gradients of the following are zero**:
 - $\nabla_{\theta} \rho_0(s_0) = 0$
 - $\nabla_{\theta} P(s_{t+1}|s_t, a_t) = 0$
 - $\nabla_{\theta} R(\tau) = 0$

Policy Gradient

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Policy Gradient

- Grad-Log-Prob of a Trajectory

$$\begin{aligned} & \nabla_{\theta} \log(P(\tau|\theta)) \\ &= \cancel{\nabla_{\theta} \log(\rho_0(s_0))} + \sum_{t=0}^T (\cancel{\nabla_{\theta} \log(P(s_{t+1}|s_t, a_t))} + \nabla_{\theta} \log(\pi_{\theta}(a_t|s_t))) \end{aligned}$$

$$\nabla_{\theta} \log(P(\tau|\theta)) = \sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t|s_t))$$

Policy Gradient

- Let 's derive the policy gradient

$$\nabla_{\theta} J(\pi_{\theta}) = \nabla_{\theta} E_{\tau \sim \pi_{\theta}}[R(\tau)]$$

$$\nabla_{\theta} J(\pi_{\theta}) = \nabla_{\theta} \int_{\tau} P(\tau|\theta) \cdot R(\tau) \quad \text{Expand expectation}$$

$$\nabla_{\theta} J(\pi_{\theta}) = \int_{\tau} \nabla_{\theta} P(\tau|\theta) \cdot R(\tau) \quad \text{Bring gradient under integral}$$

Policy Gradient

- Let 's derive the policy gradient

$$\nabla_{\theta} J(\pi_{\theta}) = \int_{\tau} P(\tau|\theta) \nabla_{\theta} \log(P(\tau|\theta)) \cdot R(\tau)$$

Log-derivative trick

$$\nabla_{\theta} J(\pi_{\theta}) = E_{\tau \sim \pi_{\theta}} [\nabla_{\theta} \log(P(\tau|\theta)) \cdot R(\tau)]$$

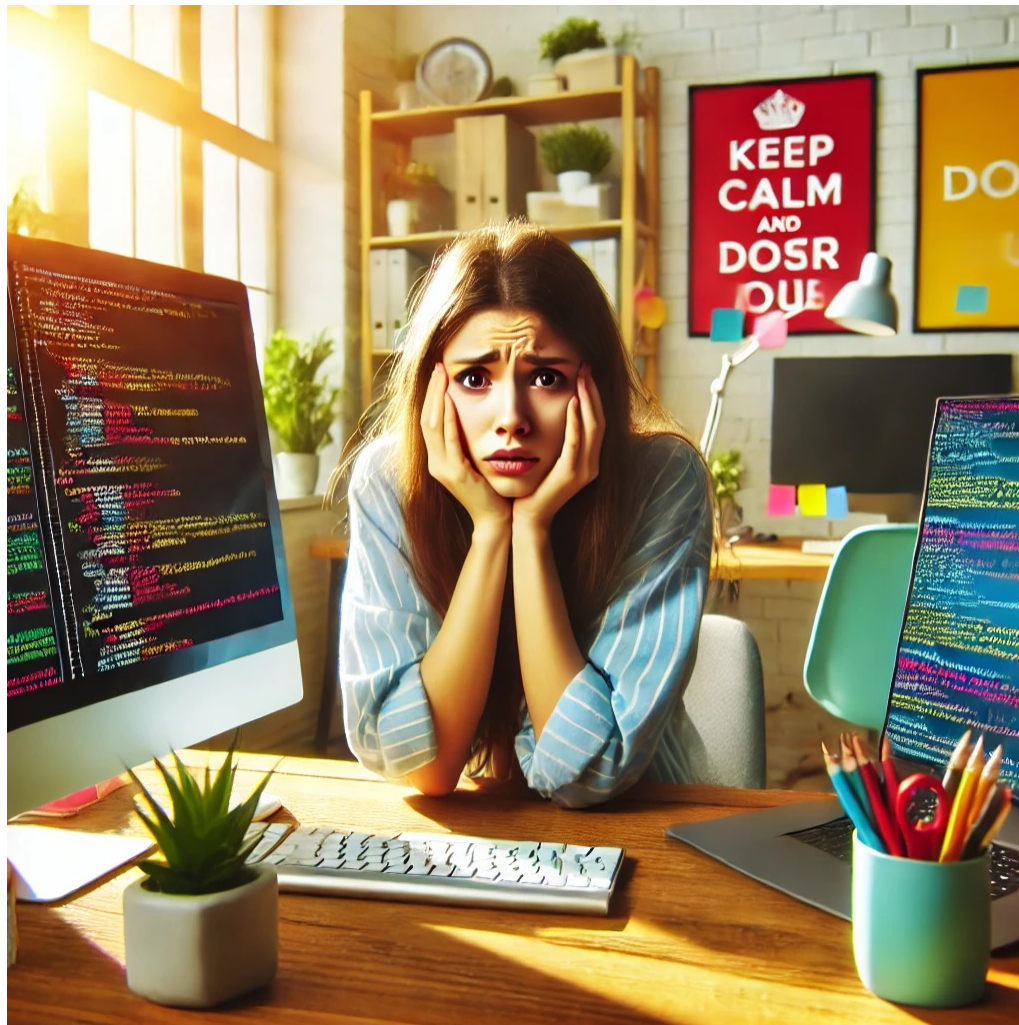
Return to expectation form

$$\nabla_{\theta} J(\pi_{\theta}) = E_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t|s_t)) R(\tau) \right]$$

Using grad-log-prob

Policy Gradient

And how do I implement this? HELP!



Policy Gradient

$$\nabla_{\theta} J(\pi_{\theta}) = E_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T R(\tau) \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) \right]$$

- We can **estimate this expected value** above with a sample mean:

- **Collect a set of trajectories:**

Collect $D = \{\tau_i\}_{i=1, \dots, N}$ using the policy π_{θ}

- **Estimate the policy gradient** as follows:

$$\hat{g} = \frac{1}{|D|} \sum_{\tau \in D} R(\tau) \sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t))$$

Thank You



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